



Final Project Demonstrations

CEN429 Secure Programming — Week 15

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This Week: Final Project Demonstrations (RAP2)

- No lecture; final report (**the complete guide**) + protected application + test results
- RAP2 = **70%** of the final · Quiz-2 = **30%** · Course grade = $0.4 \cdot \text{Grade_Midterm} + 0.6 \cdot \text{Grade_Final}$
- Detailed rubric: project guide · order and timing: in class
-  No late submissions · a **finding-action list** for the midterm feedback

RAP2 Submission — Diagram

RAP2 (final) teslimi: ne bekleniyor?

vizedeki yapının üzerine ölçüm ve kanıt eklenir

Tam ürün	C++ konsol + DLL/.so, CMake/CTest
S6–S15	kimlik, veri, kriptoloji, RASP, gizleme, hat
S16 · SONUÇLAR	plan değil, GÖZLENEN test sonuçları
S17 · Uyum matrisi	her satır kanıtlı
S14 · Devredilenler	gerekçeli, onaylı

RAP2 = 0,7 · final notu. En sık kayıp: S16'da sonuç yerine plan yazmak.

Final Rubric

Criterion	LO	Guide
Cryptography	2	S8
Secure communication	4	S6 · S11
Asset management	5	S5 (complete) · S8
Binary protections	3	S9 (advanced) · S15
Security testing and unit tests	6	S16 results
Standards	7	S1 · S14 · S17
Final report and presentation	7	All

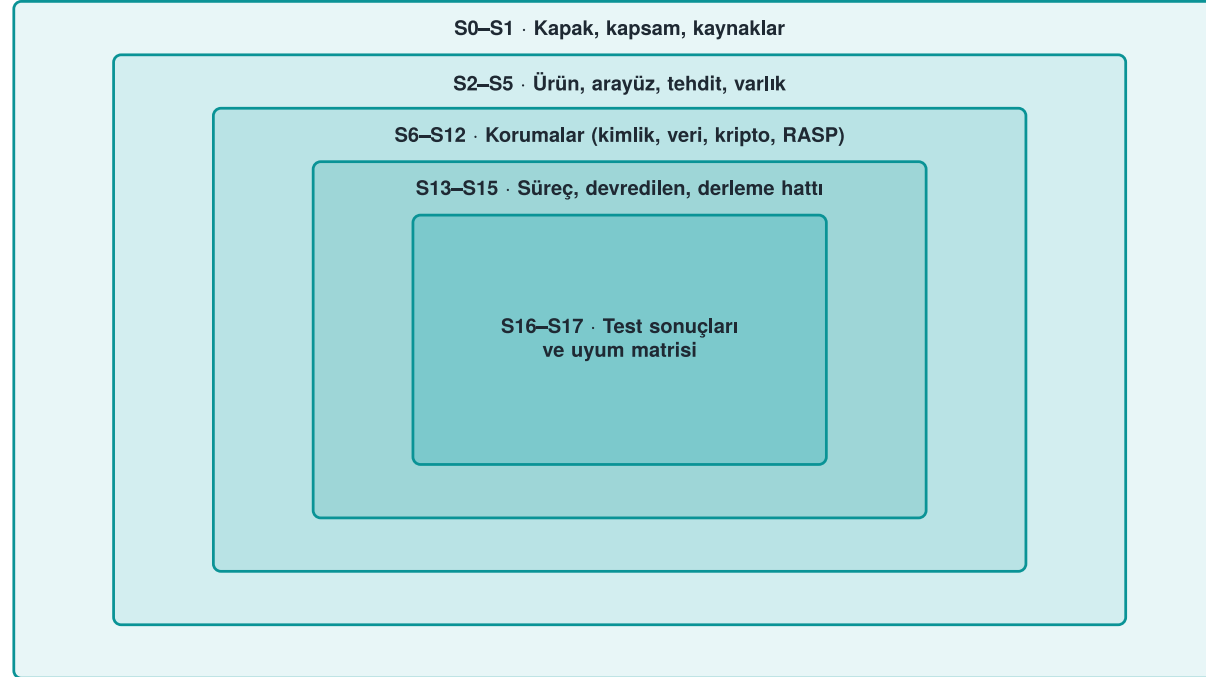
From Midterm to Final: New and Completed Sections

New	To be completed / updated
S6 identity and binding	S5 assets (complete)
S8 cryptography and keys	S9 hardening (advanced)
S11 secure communication	S12 reporting (complete)
S14 assumptions and deferred requirements	S13 updated SBOM
S15 build and deployment pipeline	S16 results · S17 complete matrix

Security Guide Map S0–S17 — Diagram

Güvenlik kılavuzu S0–S17

dıştan içe değil, alttan üste: her bölüm bir öncekine dayanır



Kılavuz, sertifikasyon belgesinin küçültülmüş modelidir.

Preparation Checklist

Week	What you should be able to show
9, 14	Obfuscated functions and justification · cost table · obfuscation and signing in the build pipeline
10	Algorithm inventory · key lifecycle · TLS/pinning · signature verification
11	Protection method or justification for a sensitive asset
12	Test plan + results · unit tests · CI · finding–action list
13	Compliance matrix · assumptions and deferred requirements

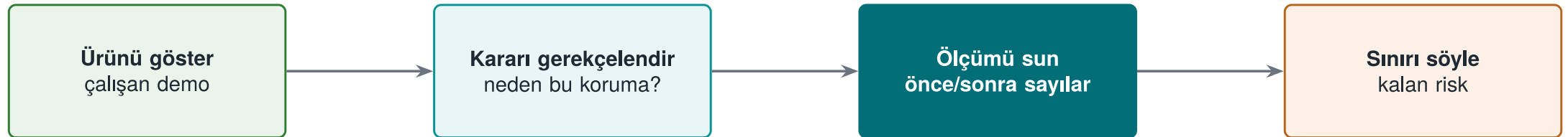
Suggested Demonstration Flow

1. **Summary:** product, architecture, the three most critical assets
2. **Since the midterm:** finding–action list (finding → action → closure; the assessor looks for this loop)
3. **Live demonstration:** cryptography, secure communication, a protection layer **working live** (e.g., a tampered file being rejected)
4. **Evidence:** tests, protection table, SBOM, compliance matrix
5. **Remaining risk:** what was deliberately scoped out, and why

Final Demonstration — Diagram

Final gösteriminin dört adımı

değerlendirici bu dördünü arar



Sınırını dürüstçe söyleyen takım, her şeyi 'güvenli' diyen takımdan yüksek alır.

Sample Questions

- Which key is this key derived from, where does it live, when is it deleted?
- How do you verify the certificate chain? Where is hostname verification performed?
- What happens if signature verification fails? Can an old but validly signed version still be installed?
- Did you measure the cost of obfuscation? Which function did you obfuscate, and why?
- In the compliance matrix, where is the evidence for this row marked "met"?
- Which requirement did you defer, and how will the other party meet it?

Common Mistakes

Mistake	Why is it a problem?
A test plan exists, but no results	At the final, results are expected
"Met" without evidence	Counted as unmet
Guide and code version differ	The assessed product becomes ambiguous
Empty remaining-risk section	Incomplete analysis
Midterm feedback was ignored	The finding–action loop is part of the process
Real secret / personal data in the repo	Values must be synthetic

Academic Integrity

The syllabus's "Academic Integrity" section also applies to the project; every submitted line must be explainable, and any team member may be asked a question during the demonstration.

S16: plan mı, sonuç mu?

finalde S16'dan GÖZLENEN sonuç beklenir

PLAN (yetersiz)

“Şu testleri yapacağız”
“Şu araçları kullanacağız”
→ hiçbir şey kanıtlanmadı

SONUÇ (beklenen)

“Şu test çalıştırıldı”
“Gözlenen çıktı şu”
“Karar: geçti / kaldı”
→ kanıt var

Test kartı: amaç · ön koşul · adımlar · beklenen · GÖZLENEN · karar · kanıt.

At the End of the Term

- Your guide = a small model of certification documents → **portfolio** (make sure it contains no confidential information)

At the final, **results** are expected: every protection is shown working, every requirement is tied to evidence.



Next Week

Week 16 — Quiz-2

After the final demonstration (RAP2), it's time for Quiz-2; the scope is weeks 9–14. The study guide is on the Week 16 page.